



TAKUMI

Expert knowledge, captured. Craftsmanship, inherited.

Preserving & Transferring Tacit Technical Expertise to Future Generations

JETRO-PS-03 · Thapar Japan Hackathon 2026 — Bridging Tradition and Innovation

Team [NAME] · [Member 1] · [Member 2] · [Member 3] · [Member 4]

trigger 0.40 V (rise) — TRIGGERED · LOCKED

CAPTURE ▶ EXTRACT ▶ REVIEW ▶ APPROVE ▶ PRACTICE

OFFLINE-CAPABLE · RECORDINGS STAY LOCAL · ZERO PAID API

The knowledge leaves before it is written down

Japan's master technicians are retiring faster than apprenticeships can replace them. What they know lives in hands and judgement — not manuals. A retiring expert takes decades of micro-decisions with them.

What a single 3-minute demonstration actually contains

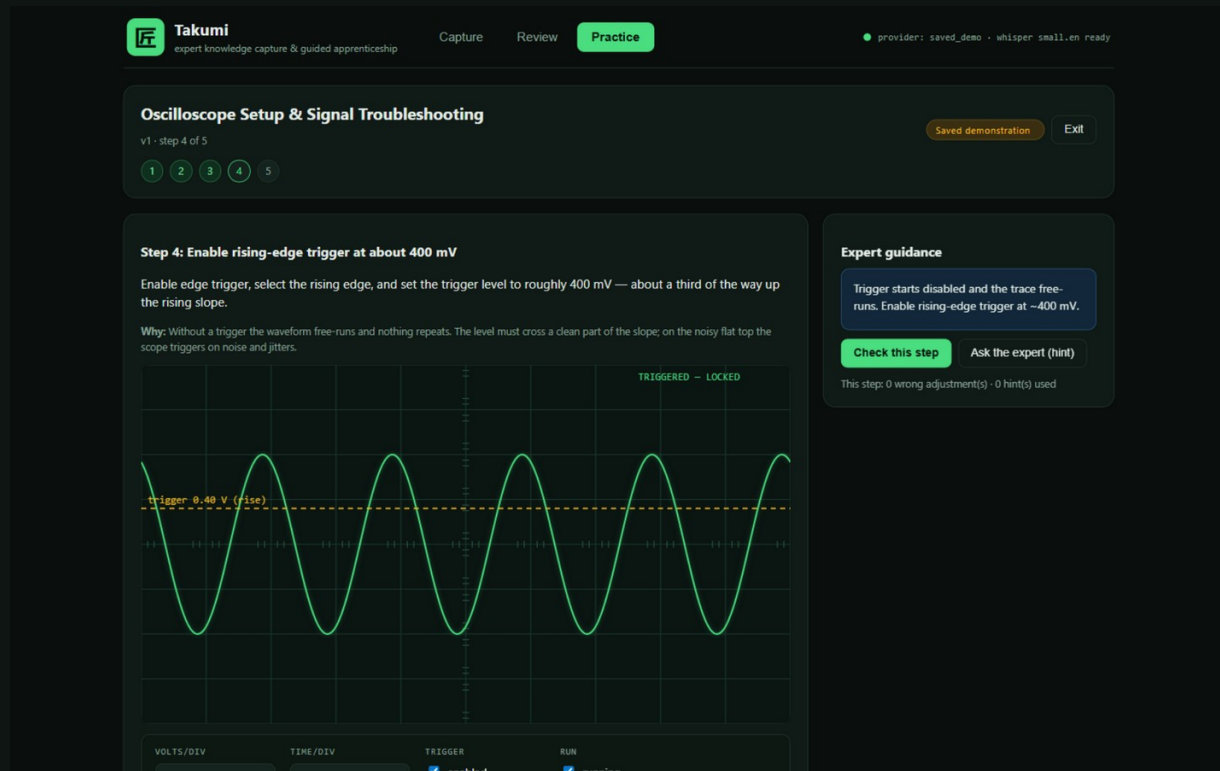
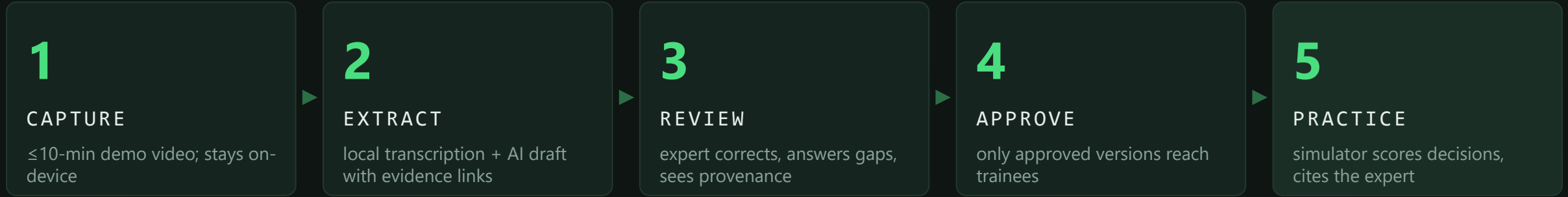
- **Rules never written** *"Connect the ground clip first — tip-first can float and damage the front end."*
- **The math behind the feel** *"500 mV/div — 2 V needs 4 of 8 divisions, leaving headroom for overshoot."*
- **Rules of thumb** *"Two cycles on screen — enough to judge period and symmetry."*
- **Failure diagnosis** *"Park the level on the noisy flat top and the display jitters all day."*
- **Recovery moves** *"Enable trigger, walk the level to a clean rising crossing, stop."*

WHY VIDEO ALONE FAILS

- Unstructured — a 5-minute video holds ~40 micro-decisions nobody can search
- Unverified — transcripts mis-hear; no link from claim to evidence
- Untested — watching is not doing; nothing checks the trainee
- Unowned — the expert never confirms what AI inferred

Evidence: bundled expert demonstration on oscilloscope setup & signal troubleshooting (5 steps extracted, 3 gaps flagged for the expert).

One video becomes a lesson a trainee can practice



DESIGN PRINCIPLES

- The expert stays the author — nothing trains without approval
- Every claim links to a verbatim transcript moment
- Missing knowledge stays visibly incomplete, never invented
- Practice checks decisions; bench observation stays human

Live prototype — feedback quotes the expert's own recovery rule.

Built like an instrument: local by default

RECORDING

video/audio · local disk, never uploaded

TRANSCRIBE

faster-whisper small.en · CPU INT8 · on-device

EXTRACT (provider adapter)

structured JSON + verbatim evidence refs

VALIDATE

schema + evidence-vs-transcript check

EXTRACT PROVIDERS

NVIDIA NIM (gated) · Ollama local · bundled offline

LESSON STORE

SQLite · draft/approved versions · attempts

REVIEW & APPROVE

expert editor; approval gating; version history

PRACTICE

scope simulator · server-recomputed scores

STACK

- React + TypeScript + Vite — canvas oscilloscope simulator
- FastAPI + SQLite (WAL) — background jobs with progress & failure states
- faster-whisper INT8 on CPU — extraction caching, bounded LLM requests
- Single-command offline demo — frontend served by the backend

GUARANTEES ENFORCED IN CODE

- Recordings stay on-device; only transcript text reaches a hosted model, if enabled
- No silent switching to paid providers — auto only steps down: hosted → local → offline
- Unapproved lessons invisible to trainees — 403 enforced server-side
- Hallucinated citations rejected — quotes must match the transcript

Verified the way engineers verify — not slideware

16/16

automated end-to-end checks
passing

100%

offline demo capability — zero
network, zero keys

0

paid API calls in the default
configuration

5

guided simulator steps incl. a
recovery scenario

What the automated suite proves

- Trainee cannot fetch an unapproved lesson — 403, server-enforced
- Out-of-range or non-matching evidence references rejected before review
- Approval blocked while any follow-up question is unresolved
- Attempt scores recomputed server-side — client cannot fake a 100
- Data survives restart; interrupted jobs marked failed, never stuck

What we ran live

- Full browser walkthrough: capture → review (answers + approval) → practice (all 5 steps) → attempt stored
- Wrong adjustment penalised exactly as designed (95/100 with one mistake)
- Provider failure paths: malformed AI output, dead provider, oversized upload — all fail loudly with guidance
- Health & provider readiness surfaced in the UI (hosted/local/offline provenance)

Planned effectiveness study (disclosed, small-sample): trainees with Takumi vs. conventional written instructions — measure completion time, mistakes, assistance needed. No effectiveness claims until measured.

Source: Takumi test suite & verification log, docs/VERIFICATION.md, October 2026.

Where Takumi goes after the bench

Business hypothesis — to be validated with customers, not assumed

Organisation subscription for teams that must inherit skills: Japanese SMEs onboarding an Indian workforce, maintenance & vocational training institutes, and manufacturing partners — the exact corridor JETRO serves. Pilots would measure time-to-competence before any pricing claim.

WHO PAYS

Factories, training institutes, service organisations with retiring experts and rotating trainees.

WHY US

Evidence-verified lessons, expert-in-the-loop approval, offline-capable — built for shop floors, not just offices.

PILOT PLAN

3 departments · 2 lessons each · measure time-to-competence and mistakes vs. current handover method.

THE ASK

- One JETRO-introduced factory or training partner for a real pilot recording
- Mentor feedback on the evidence-validation approach for industrial know-how

18 OCT Round-1 submission **24–25 OCT** live build & pitch — demo runs offline from one command

⌘ Takumi — bridging tradition and innovation, one demonstration at a time.